

Sign of permutation.

Let a polynomial over $\{\pm 1\}$ with distinct n -variables $x_1, \dots, x_n$ as

$$\Delta \stackrel{\text{def}}{=} \prod_{1 \leq i < j \leq n} (x_i - x_j)$$

For each $\sigma \in S_n$, let σ act on Δ by permuting:

$$\sigma(\Delta) = \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)})$$

Since σ is bijective, there are two possibilities:

$$\sigma(\Delta) = \pm \Delta.$$

Now, let $\text{sign}: S_n \rightarrow \{\pm 1\}: \sigma \mapsto \begin{cases} 1 & \text{if } \sigma(\Delta) = \Delta \\ -1 & \text{if } \sigma(\Delta) = -\Delta \end{cases}$.

Prop. Sign is a surjective homomorphism.

Proof. Given $\sigma, \tau \in S_n$,

$$(\tau \circ \sigma)(\Delta) = \prod_{1 \leq i < j \leq n} (x_{(\tau \circ \sigma)(i)} - x_{(\tau \circ \sigma)(j)})$$

Suppose $\sigma(\Delta)$ has exactly k factors of $(x_i - x_j)$ with $j > i$. That is, $\text{sign } \sigma = (-1)^k$.

$$\text{So, } \sigma(\Delta) = \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)}) = \text{sign } \sigma \cdot \prod_{1 \leq p < q \leq n} (x_p - x_q)$$

Therefore, $(\tau \circ \sigma)(\Delta) = \text{sign } \sigma \cdot \prod_{1 \leq p < q \leq n} (x_{\tau(p)} - x_{\tau(q)})$, it follows

$$\text{sign } (\tau \circ \sigma) = \text{sign } \sigma \cdot \text{sign } \tau.$$

Surjectiveness: clearly, $\text{sign}(id) = 1$. And, $\text{sign}((12)) = -1$, so clear. $\square$

Prop. $\ker \text{sign} = A_n$.

Prop. m -cycle is odd iff m is even

Prop. a permutation σ is odd iff number of cycles of even length is odd, in decomposition.

[Prop. S_n is generated by $\{(i \ i+1) \mid 1 \leq i \leq n-1\}$.

Proof. Given $j \in \{2, \dots, n-1\}$,
 $(1 \ 2 \ 3 \ \dots \ n) = (1 \ n)(1 \ n-1) \dots (1 \ 3)(1 \ 2)$
 $(1 \ 2)(2 \ 3) \dots (j-1 \ j) = (1 \ 2 \ 3 \ \dots \ j)$

So, the generated set contains $(1 \ 2 \ \dots \ j)$ for all $2 \leq j \leq n$.

Moreover, observe that

$$(2 \ 3)(1 \ 2)(2 \ 3)^{-1} = (2 \ 3)(1 \ 2 \ 3) = (1 \ 3)$$

$$(3 \ 4)(1 \ 3)(3 \ 4) = (3 \ 4)(1 \ 3 \ 4) = (1 \ 4)$$

$$\vdots$$

$$(j \ j+1)(1 \ j)(j \ j+1) = (1 \ j)$$

Now, for any $1 \leq j < k \leq n$,

$$(j \ k) = (1 \ k)(1 \ j)$$

can be generated by the given set. All permutations can be represented by product of transpositions, so $S_n = \langle (i \ i+1) \mid 1 \leq i \leq n-1 \rangle$

[Corollary, $S_n = \langle (1 \ 2), (1 \ 2 \ \dots \ n) \rangle$

Proof. observe that

$$(1 \ 2 \ \dots \ n)(1 \ 2)(1 \ 2 \ \dots \ n)^{-1} = (1 \ 2 \ \dots \ n)(1 \ 2)(n \ n-1 \ \dots \ 1) = (2 \ 3)$$

$$(1 \ 2 \ \dots \ n)(2 \ 3)(1 \ 2 \ \dots \ n)^{-1} = (3 \ 4)$$

$\vdots$

So, $\langle (1 \ 2), (1 \ 2 \ \dots \ n) \rangle$ contains $(i \ i+1)$ for all $1 \leq i \leq n-1$, proved

+ Note that

$$(a_1 \ a_2 \ \dots \ a_n)$$

$$= (a_1 \ a_n)(a_1 \ a_2 \ \dots \ a_{n-1})$$

$$= (a_1 \ a_n)(a_1 \ a_{n-1})(a_1 \ a_2 \ \dots \ a_{n-2})$$

$$\vdots$$

$$= (a_1 \ a_n)(a_1 \ a_{n-1}) \dots (a_1 \ a_2)$$

Thm. For a prime number p , S_p is generated by any one transposition and p -cycle.
 Proof. WLOG, let the one transposition is $(1\ x)$ and p -cycle is $(1\ \dots\ p)$.
 where $x \in \{2, \dots, p\}$. And, since p is prime number,

$$(1\ 2\ \dots\ p)^k \quad (1 \leq k \leq p-1)$$

are all p -cycles. And, take $k = x-1$, then

$$(1\ 2\ \dots\ p)^k = (1\ x\ \dots)$$

$$\text{So } \langle (1\ x), (1\ \dots\ p) \rangle = \langle (1\ x), (1\ x\ \dots) \rangle = S_p.$$

Note: $\langle (1\ 3), (1\ 2\ 3\ 4) \rangle$ cannot generate S_4 . Observe that

$$(1\ 3)^2 = \text{id}, \quad (1\ 2\ 3\ 4)^2 = (1\ 3)(2\ 4), \quad (1\ 2\ 3\ 4)^3 = (1\ 4\ 3\ 2)$$

$$(1\ 3)(1\ 2\ 3\ 4) = (1\ 2)(3\ 4)$$

$$(1\ 3)(1\ 2\ 3\ 4)^2 = (2\ 4)$$

$$(1\ 3)(1\ 2\ 3\ 4)^3 = (1\ 4)(2\ 3)$$

$$\text{So, } \langle (1\ 3), (1\ 2\ 3\ 4) \rangle = \langle (1\ 2)(3\ 4), (1\ 3)(2\ 4) \rangle.$$

Prop. A_4 has no subgroup of order 6.

Proof. Let $V = \{id, (12)(34), (13)(24), (14)(23)\}$ be a subset of A_4 .

Except $id \in V$, the elements of V have order of 2, exactly, these are all elements of order 2 in A_4 .

The Conjugate cannot change the order and parity, V is a normal subgroup of A_4 .

And, since A_4/V has order of 3, it must be contain order 3 element, so, it must be isomorphic to $\mathbb{Z}_3$, cyclic.

Meanwhile, if there exists a order of 6 subgroup, N , then since $[A_4 : N] = 2$, thus for any $g \in A_4$, $gN = Ng$. so it is normal.

Now, $A_4/N \cong \mathbb{Z}_2$, cyclic.

Meanwhile, observe that: for $G' = \{x^{-1}y^{-1}xy \mid x, y \in G\}$, and $N \trianglelefteq G$,

G/N is abelian iff $G' \leq N$

To show this, let $x^{-1}y^{-1}xy \in G'$ be given. Since G/N is abelian,

$$x^{-1}y^{-1}xyN = x^{-1}N y^{-1}N xN yN = x^{-1}N xN y^{-1}N yN = N$$

so $x^{-1}y^{-1}xy \in N$.

Therefore, $G' \leq N$ and $G' \leq V$. And, since A_4 is not abelian, $G' \neq 1$.

And, if $|G'| = 2$, then it must be contain an order of 2 element, but since G' is normal, then it has all order of 2 element.

Thus, $G' \geq V$, or $G' = V$. Now, $|G'| = 4$ but $G' < N$, it is contradiction to the Lagrange Thm.